GLENCORE

Master Data Standard

Functional Location

12

Document properties

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1. Introduction

1.1 Master Data Standards - Purpose

The suite of Master Data Standards provide clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Functional Location Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for a SAP FLOC Master record at Glencore.
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5
- Identify the minimum requirements for creating, maintaining, and retiring a Functional Location
- Provide the guiding principles to manage Functional Location Master Data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is is targeted at the following audience:

- Master data maintenance teams in order to understand how the object needs to be maintained
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to: Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard
- Transition to support teams to assist in the developing of user guides

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders

2. Data Object Description

2.1 Definition

A Functional Location is part of a multi-level hierarchical structure that may include Equipment, Assemblies (Construction Type/Sub Assemblies) and Bills of Material (BOMs) organised according to spatial, technical or functional criteria. Functional Locations will represent a location within a business where a piece of plant is located or a fleet operates.

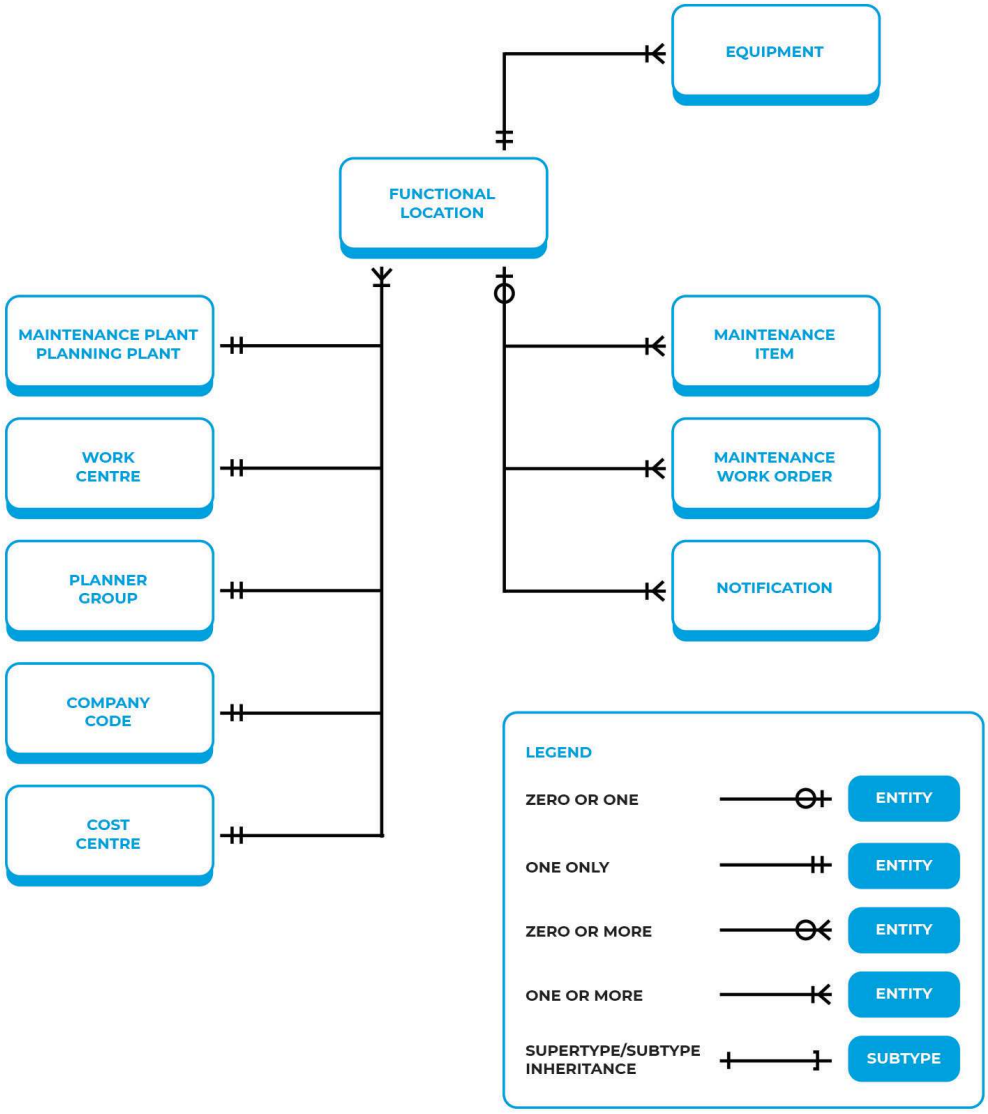
A Functional Location is created for the purpose of:

- Representing the operating system according to site, area, function and processes
- Linking Equipment
- Collecting technical data for the individual levels of the structure for evaluation
- Monitoring the costs of all maintenance tasks for the individual levels of the structure

To assist usability, Functional Locations are constructed into a single navigational hierarchy cascaded in 5 logical groupings:

- Operation
- Core Process
- Main Process
- Process
- Sub Process

2.2 Entity Relationship Diagram (ERD)



3. Business Requirements

3.1 Business design Rules & Criteria

Functional Locations at Glencore will comprise of one 'navigational' structure.

Navigational Structure – Used to define the layout and aid navigation at a particular site. The highest level of the Navigational Structure is the operation. The lowest level of the Navigational Structure will be a Functional Location representing the sub-process.

Key characteristics of the Navigational Structure:

- Defines the Operating Site
- Identifies an area within the Site
- Identifies the Operational Process and Sub-process
- Identifies key functions within the process
- Notifications or Work Orders can only be created against Functional Locations in the Navigational Structure for circuit maintenance or collective labour service orders. If required, it is preferred that Notifications and Work Orders may be created against Functional Locations in the Navigational Structure at Level 5.

In addition to the considerations described above, the following criteria will also apply:

- One edit mask (structure indicators) will be used to represent the Navigation structure. Within this single edit mask, the Navigational structure is clearly identifiable. Refer to section 4.2.
- All 5 levels of the functional location structure will be used to determine the configuration of the operation or asset.

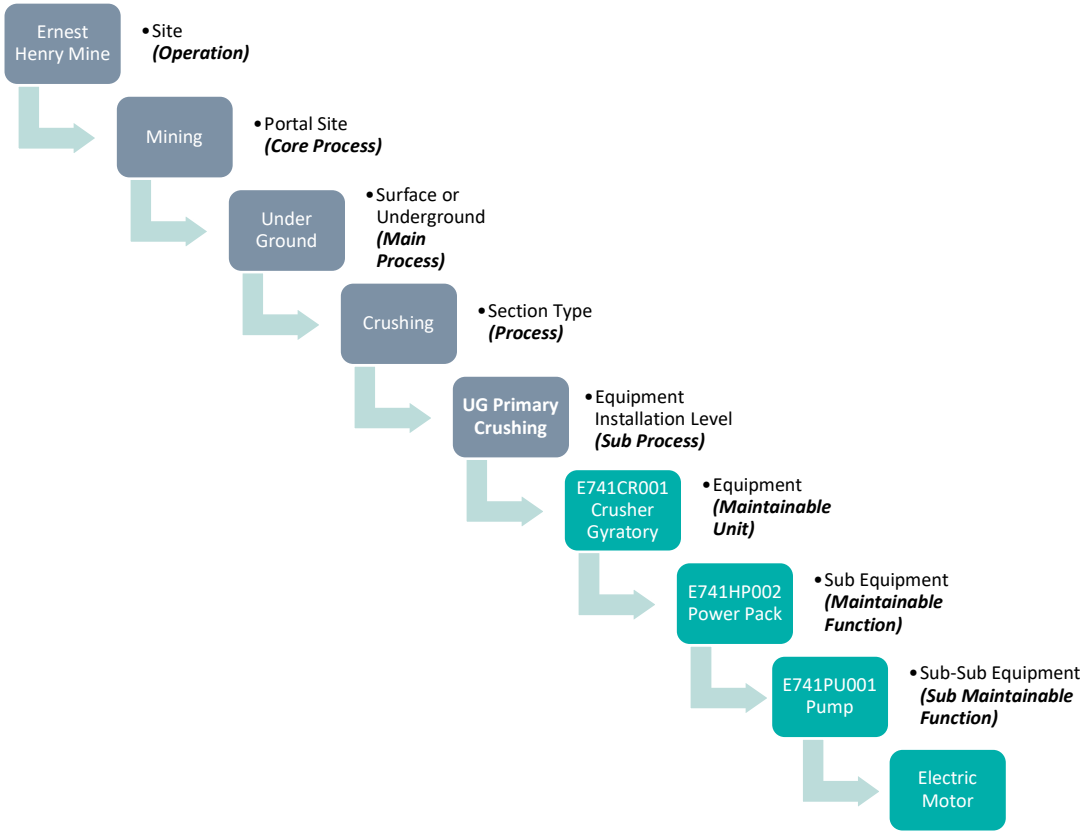
Financial Asset linkage

- The Cost Centre assigned to the functional location will be linked to the equivalent Equipment Operating (Opex) Cost Centre using the financial asset.

3.2 Hierarchy Requirements

To provide a hierarchy to locate and report Equipment within a site. This hierarchy will be used for navigation only, except where circuit maintenance is executed at level five. The Navigational Structure will have five levels.

Example of Functional Location & Equipment Structure



4. Detail Standard

4.1 Overview

A Functional Location is a data object used to define the hierarchy structure as well as the individual objects within that structure.

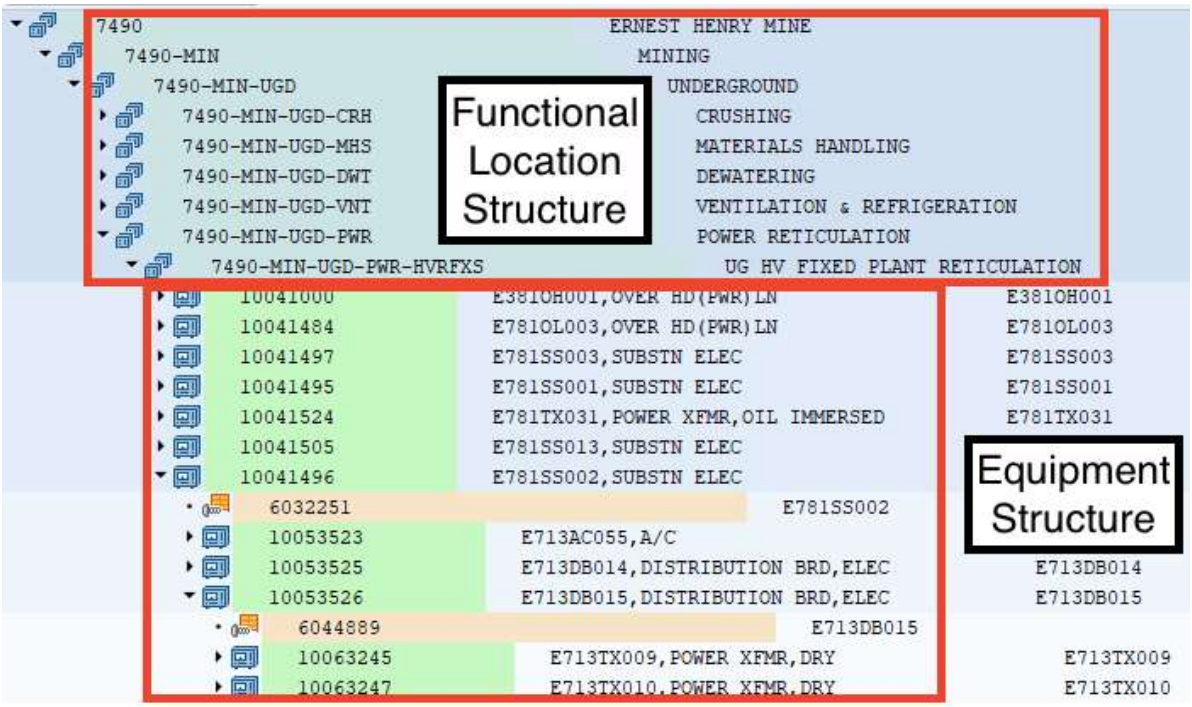
4.2 Data Object Coding Standard

Each Functional Location will be unique across the organisation and once created the code is not able to be changed or reused.

At Glencore Copper the agreed mask will be defined as follows:

- XXXX-XXX-XXX-XXX-XXXXXX representing Operation, Core Process, Main Process, Process & Sub Process

  1 - 2 - 3 - 4 - 5 ← Hierarchy Levels



Functional Locations that are no longer required (i.e. made redundant, decommissioned) will be shifted to a Graveyard Functional Location that is local to each Operation. The agreed mask is defined as follows:

- XXXX-DEL-DEL-DEL for each Operation (XXXX)

4.3 Attributes

4.3.1 Functional Location Code

Application Definition	The code used to identify unique Functional Locations within the technical Object Structure
Guiding Principles	Functional location code must be unique for all items across all operations. Uniqueness will be achieved by adding the SAP Maintenance Plant Code to the top level of the Structure.
Application Coding Convention	40 characters – Alphanumeric.
Naming Convention	Refer to section 4.2 and MDD General Configuration Standard
Reporting Considerations	The mask FLOC codes provide the back bone for recording maintenance history and cost against assets

4.3.2 Functional Location Short Text

Application Definition	Text used to describe the Functional Location
Guiding Principles	Short text will conform to the Data Standard for short text descriptions and will describe the plant item being maintained
Application Coding Convention	40 Characters – Alphanumeric
Naming Convention	Refer to MDD General Configuration Standard
Reporting Considerations	Will support all reports pertaining to Maintenance, History and Costing

4.3.3 Functional Location - Superior Functional Location

Application Definition	The code used to identify unique Superior Functional Locations within the technical Object Structure
Application Guiding Principles	The Superior Functional Location automatically defaults to the next level up Functional Location with the same structure indicator.
Guiding Principles	The Superior Functional Location is used to provide a Parent Child Hierarchical structure.
Application Coding Convention	As per Functional Location.
Naming Convention	Refer to section 4.2 & MDD General Configuration Standard

4.3.4 Functional Location – Cost Centre

Application Definition	Key that uniquely identifies a cost centre.
Application Guiding Principles	The number is externally assigned and can be alphanumeric.
Guiding Principles	The cost centre in a functional location will be populated with an Equipment Maintenance Cost Centre. This Cost Centre collects maintenance costs by type of equipment. The costs on the Equipment Maintenance Cost Centre will be allocated to the corresponding Equipment Operating (Opex) Cost Centre to allow reporting and analysis of the total operating costs for the physical assets at a site.
Application Coding Convention	10 character alphanumeric code
Naming Convention	Glencore will use a 10 character (numeric) Cost Centre coding convention. MDD General Configuration Standard
Reporting Considerations	The Cost centre number is a critical attribute for reporting purposes. It is a key parameter for reporting and will be extracted from SAP ERP as part of core master data to the data reporting tool

5. Appendix A: Additional Information

5.1 Complete Field List

Legend for Attribute Table

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, DATS=Date, CUKY=Currency key, UNIT=Measurement type, NUMC=Number)
- **Length –** Character length of field
- **MOID –** Data requirement (M=Mandatory, O=Optional, I=Internally populated, D=Data Origin i.e. Inherited from the Superior Functional Location)
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

Master Data Standard - Functional Location

Table	Field	Field Name	Description	Data Type	Len	MO I D	Field Validation Type
IFLO	TPLNR	Functional Location	Label that uniquely identifies a Functional Location	CHAR	40	M	23 Character Max at Glencore. Refer to data coding standard in section 4.2 & MDD General Configuration Standard
IFLO	FLTYP	Functional Location category	Key which shows the functional location category	CHAR	1	M	Table select: M - Technical system – standard
IFLO	PLTXT	Description of Functional Location	Description of the functional location in the form of short text	CHAR	40	M	Non-validated 40 char field – refer to Data Standard for short texts.
ITOB	SWERK	Maintenance plant	Number which clearly identifies a maintenance plant.	CHAR	4	M	Refer to General Configuration Standrad
ITOB	BEBER	Plant section	Number which clearly identifies a plant section. The table values are maintenance plant specific and may be used to group into maintenance business/sub-business units within a plant	CHAR	3	M	Table Select: 3 Character alphanumeric.
ITOB	BUKRS	Company Code	The Company code is an organisational unit within financial accounting	CHAR	4	D	Value defaults from maintenance plant
ITOB	KOSTL	Cost Centre	Key uniquely identifying a cost centre – the default order settlement cost object when a work order is created from the Functional Location. Normally operating costs (OPEX)	CHAR	10	M	Each maintainable Functional Location must have a cost object of cost centre. Refer to Functional Location – Cost Centre in the attributes section.
ITOB	IWERK	Planning Plant	Number which clearly identifies the plant at which the planning function is carried out.	CHAR	4	M	Table select: Valid table entry. At Glencore this will match the Maintenance plant

Table	Field	Field Name	Description	Data Type	Len	MO I D	Field Validation Type
ITOB	INGRP	Planner Group	Key that uniquely identifies the planner group within the planning plant responsible for the Functional Location	CHAR	3	M	Table Select: 3 Character alphanumeric.
ITOBATTR	GEWRK	Main work centre	Key that uniquely identifies the main work centre within the site, responsible for maintenance tasks on the Functional Location	CHAR	6	M	Mandatory for all Functional Location levels.
ITOBATTR	TPLKZ	Structure Indicator	The structure indicator of a functional location determines the structure of the functional location label. It establishes the following factors: • Number of hierarchy levels of a technical system • Key length of the levels • Separators (hyphens)	CHAR	4	I	Z001 – Glencore Copper Structure Indicator
ITOB	TPLMA	Superior Functional Location	Functional Location which is hierarchically directly above the Functional Location that you are creating, changing or displaying. If the system can determine a superior Functional Location on the basis of the structure that you are using, it will enter its label as a default value in this field when you create a new Functional Location.	CHAR	40	M	Mandatory except for top level Functional Location.
ITOB	POSNR	Position in superior technical object	The default position will be alphabetical or numerical order when displayed using IH01. To modify the sequence, enter position numbers for all objects at the same level. The system does not check the content of the field Position. Particularly in the case of Functional Locations, it does not check whether position labels are unique under a superior Functional Location.	CHAR	4	O	Free text 4 character field.
ITOBATTR	IEQUI	Inst of Equipment Allowed at Functional Location	If selected allows Equipment to be Installed at the Functional Location	CHAR	1	O	Whilst this is an optional selection, the Enabled (Ticked) condition shall be set as a default.

Table	Field	Field Name	Description	Data Type	Len	MOID	Field Validation Type
							Must be unticked for levels 1 to 4.
ITOBATTR	EINZL	Single equipment installation at Functional Location	Indicator that shows whether only single equipment or several pieces of equipment can be installed at a Functional Location at the same time. If you wish to allow the installation of only a single piece of equipment at this Functional Location, flag this field	CHAR	1	O	All Functional Locations shall allow for multiple installations.

6. Stakeholder Signoff

Stakeholder Name	Role	Contact	Stakeholder Sign-off
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